
A Unified Empirical Study and Optimization of Data-Free Model Merging under Physical Quantization and Real-World Corruptions

Pragmatic Researcher¹

Abstract

Merging multiple specialized neural networks into a single multi-task model has emerged as a powerful training-free approach to consolidate skills. However, existing work almost exclusively focuses on the clean, full-precision (FP32) performance of merged models. In contrast, real-world deployment on edge devices is constrained by strict resource limits, requiring post-training quantization (PTQ) to low-bit formats (e.g., 8-bit or 4-bit) and robustness to environmental corruptions like sensor noise and blur. In this paper, we conduct the first systematic, unified empirical investigation of data-free model merging under these physical constraints. We show that while parameter-space calibration methods like Isotropic Parameter Resonance (U-IPR) and Holographic Norm Scaling (HNS) successfully restore representation scaling in full precision, their calculated scale factors shift the dynamic range of weight updates, severely amplifying quantization noise and sensitivity under out-of-distribution corruptions. To resolve this, we propose Quantization-Robust Parameter Resonance (QR-IPR), which dynamically clamps scale factors using robust layer-wise statistics (Median and Median Absolute Deviation). Our comprehensive benchmark of Weight Averaging (WA), Task Arithmetic (TA), TIES-Merging, and DARE-Merging across MNIST, FashionMNIST, and CIFAR-10 ResNet-18 experts demonstrates that QR-IPR effectively stabilizes models under 8-bit quantization and noise while preserving full-

precision accuracy. Finally, we expose a critical boundary: all zero-shot parameter merging methods collapse under 4-bit uniform quantization, establishing key guidelines for practitioners.

1. Introduction

The exponential growth in the number of specialized deep neural networks fine-tuned for distinct downstream tasks has made multi-task serving a major operational bottleneck. Keeping independent model checkpoints for each task is extremely expensive in terms of storage, memory, and routing latency, particularly on edge hardware (e.g., mobile phones, autonomous vehicles, and remote sensors). To resolve this, model merging has emerged as an active area of study (Haji et al., 2024; Le et al., 2023). By combining multiple specialized models sharing a common pre-trained progenitor, practitioners can construct a unified multi-task model with zero additional training cost, zero training data, and a $1\times$ storage footprint.

Two foundational paradigms dominate parameter-space model merging: Weight Averaging (WA) (Wortsman et al., 2022) and Task Arithmetic (TA) (Ilharco et al., 2023). While conceptually elegant, simple linear merges often experience performance degradation due to interference between conflicting task-specific parameters (Yadav et al., 2023). To mitigate this, advanced sparsification and sign election methods such as TIES-Merging (Yadav et al., 2023) and DARE (Yu et al., 2024) have been developed. More recently, parameter-space calibration methods have been introduced to address representation collapse (the degradation of hidden representation scaling) by rescaling task updates. Notable examples include Holographic Norm Scaling (HNS) (Anonymous, 2026a) and Update-level Isotropic Parameter Resonance (U-IPR) (Anonymous, 2026b).

Despite their theoretical appeal, current model merging and calibration methods suffer from a severe academic-to-practical gap. Research papers evaluate

¹Pragmatic AI Systems Lab, Department of Computer Science, University of Technology. Correspondence to: Pragmatic Researcher <researcher@pragmatic-ai.edu>.

merging algorithms almost exclusively in full precision (FP32 or FP16) on clean, curated datasets. However, real-world deployment on physical edge hardware demands two major concessions: (1) Low-Bit Quantization: models are quantized to 8-bit (INT8) or 4-bit (INT4) to fit VRAM limits and leverage hardware-level accelerators. (2) Environmental Corruptions: input signals are frequently corrupted in the wild by factors like sensor noise and blur (Hendrycks & Dietterich, 2019).

In this paper, we adopt the perspective of The Pragmatist and systematically bridge this gap. We discover that data-free calibration methods like HNS and U-IPR, although highly effective at FP32, are highly sensitive to quantization and environmental noise. Specifically, we show that these methods compute channel-wise or layer-wise scaling factors that can grow excessively large. Under Post-Training Quantization (PTQ), these extreme scale factors inflate the dynamic range of weight updates, severely amplifying quantization rounding and clipping errors. Similarly, they amplify high-frequency environmental noise, leading to representation collapse in the wild.

To resolve these practical bottlenecks, we propose Quantization-Robust Parameter Resonance (QR-IPR), a simple, highly effective, and data-free calibration technique. QR-IPR utilizes robust layer-wise statistics—namely, the Median and Median Absolute Deviation (MAD) of computed scale factors—to dynamically clamp channel-wise scaling multipliers. By isolating and regularizing outlier scales, QR-IPR preserves the representative scaling benefits of parameter resonance while ensuring maximum stability under physical quantization and environmental perturbations.

We present a thorough evaluation using ResNet-18 (He et al., 2016) experts trained on MNIST (LeCun et al., 1998), FashionMNIST (Xiao et al., 2017), and CIFAR-10 (Krizhevsky & Hinton, 2009). Our main contributions are: (i) we conduct the first unified empirical study of data-free model merging under physical quantization (INT8, INT4) and environmental corruptions (noise, blur); (ii) we mathematically analyze and empirically demonstrate how standard calibration (HNS, U-IPR) degrades under quantization and noise due to outlier scale inflation; (iii) we introduce QR-IPR, a novel data-free calibration technique that dynamically clamps scale factors; (iv) we mathematically and empirically validate that update-level calibration acts as a unifying attractor, causing the Task Arithmetic scaling factor λ to cancel out; and (v) we identify a critical boundary for zero-shot PTQ merging, showing that

per-tensor 4-bit quantization causes complete collapse of all methods.

2. Related Work

Linear Mode Connectivity & Weight Averaging. Models fine-tuned from the same pre-trained initialization often lie within the same low-loss basin, a phenomenon known as Linear Mode Connectivity (LMC) (Entezari et al., 2022; Ainsworth et al., 2022). This forms the foundation for Weight Averaging (WA) and Model Soups (Wortsman et al., 2022), which average parameters directly to improve generalization.

Task Arithmetic & Advanced Merging. Task Arithmetic (TA) (Ilharco et al., 2023) defines a "task vector" as the parameter difference between the fine-tuned model and its pre-trained progenitor. These task vectors can be added or subtracted to combine or remove skills. However, merging multiple task vectors often introduces parameter interference, where updates to the same parameter conflict. Advanced methods like TIES-Merging (Yadav et al., 2023) and DARE (Yu et al., 2024) resolve this by trimming small values and selecting parameter signs via sign election.

Parameter-Space Calibration. Merging models linearly often scales activations and representations incorrectly, causing magnitude and representation collapse. REPAIR (Jordan et al., 2022) stabilizes activations using a data-driven calibration step. To eliminate the need for calibration datasets, Holographic Norm Scaling (HNS) (Anonymous, 2026a) derives analytical scaling factors from parameter statistics alone to restore task update scaling. Similarly, Isotropic Parameter Resonance (U-IPR) (Anonymous, 2026b) rescales task updates layer-wise to match the average isotropic scale of individual experts using Frobenius norms.

Post-Training Quantization (PTQ). Quantization is the standard industry practice for deploying deep learning models on resource-constrained devices (Dettmers et al., 2022; Frantar et al., 2023; Xiao et al., 2023). PTQ quantizes weights and activations without re-training. Recent studies examine the intersection of quantization and merging, such as TVQ (Kim et al., 2025) and Merge-Friendly PTQ (Shin et al., 2025). However, no prior work has systematically analyzed how parameter-space calibration methods interact with quantization noise and input corruptions.

3. Analyzing the Robustness Bottleneck of Data-Free Calibration

Let θ_{init} be the pre-trained progenitor model’s parameters. We have K expert models trained on K distinct tasks, denoted as θ_t for $t \in \{1, \dots, K\}$. The task up-

date vector for expert t is defined as:

$$\tau_t = \theta_t - \theta_{\text{init}} \quad (1)$$

Under standard Task Arithmetic (TA) with scaling factor λ , the merged model parameters θ_{merged} are:

$$\theta_{\text{merged}} = \theta_{\text{init}} + \lambda \sum_{t=1}^K \tau_t \quad (2)$$

The merged update is defined as $T_{\text{merged}} = \theta_{\text{merged}} - \theta_{\text{init}} = \lambda \sum_{t=1}^K \tau_t$.

To calibrate this update and prevent representation collapse, U-IPR (Anonymous, 2026b) rescales T_{merged} layer-wise such that its Frobenius norm matches the average Frobenius norm of individual expert updates:

$$s_{\text{U-IPR}} = \frac{\frac{1}{K} \sum_{t=1}^K \|\tau_t\|_F}{\|T_{\text{merged}}\|_F} \quad (3)$$

And the calibrated weights are $\theta_{\text{cal}} = \theta_{\text{init}} + s_{\text{U-IPR}} T_{\text{merged}}$.

HNS (Anonymous, 2026a) extends this to channel-wise scaling. For each output channel (or filter) c of a weight matrix, HNS computes:

$$s_c = \frac{\frac{1}{K} \sum_{t=1}^K \|\tau_{t,c}\|_2}{\|T_{\text{merged},c}\|_2} \quad (4)$$

And the calibrated task update is $(T_{\text{cal}})_c = s_c T_{\text{merged},c}$. To handle numerical instability and noisy channels, both U-IPR and HNS clamp their computed scale factors to a global, heuristic bounding box:

$$s_{\text{clamped}} = \text{clip}(s, s_{\min}, s_{\max}) \quad (5)$$

where $s_{\min} = 0.1$ and $s_{\max} = 10.0$ are typically chosen.

3.1. Quantization Noise Inflation

We identify that the dynamic range of weight values is a critical factor for quantization robustness. Let W be a weight tensor to be quantized. In symmetric uniform quantization, the quantized weight is:

$$W_{\text{quant}} = \text{clamp} \left(\text{round} \left(\frac{W}{\Delta} \right), -q_{\max}, q_{\max} \right) \times \Delta \quad (6)$$

where $\Delta = \frac{\max(|W|)}{q_{\max}}$ is the quantization step-size, and $q_{\max} = 2^{b-1} - 1$ for a b -bit integer.

Notice that the step-size Δ depends directly on the maximum absolute value of the weight tensor, $\max(|W|)$. If a tensor contains even a single outlier with a very large magnitude, the step-size Δ becomes

extremely large. This increases the rounding error for all other parameters in the tensor, leading to severe quantization noise and representation collapse.

When applying HNS or U-IPR, some channels or layers receive very large scaling multipliers (close to $s_{\max} = 10.0$). This directly inflates the magnitude of task updates for those specific channels or layers. If those updates contain outliers, or if the initial weights were already large, the calibrated weight values $W_{\text{cal}} = \theta_{\text{init}} + s T_{\text{merged}}$ experience a dramatic expansion in their maximum absolute magnitude. This blows up the quantization step-size Δ , causing massive rounding errors and destructive representation collapse under PTQ.

Proposition 3.1 (Per-Tensor Quantization Error Bounds). Let $W_{\text{cal}} \in \mathbb{R}^{d_o \times d_i}$ be the calibrated weight matrix of a convolutional or linear layer, represented as channel-wise vectors $W_{\text{cal},c} = W_{\text{init},c} + s_c T_{\text{merged},c}$ for output channel $c \in \{1, \dots, d_o\}$, where $s_c \geq 0$ is the scale factor and $T_{\text{merged},c}$ is the merged task update. Under b -bit symmetric uniform per-tensor quantization with $q_{\max} = 2^{b-1} - 1$, the entry-wise quantization error is bounded by:

$$\begin{aligned} \|W_{\text{cal}} - Q(W_{\text{cal}})\|_{\infty} &\leq \frac{\max_c \|W_{\text{init},c} + s_c T_{\text{merged},c}\|_{\infty}}{2(2^{b-1} - 1)} \end{aligned} \quad (7)$$

Furthermore, let the error bound under HNS be E_{HNS} and under QR-IPR be $E_{\text{QR-IPR}}$. If $s_{c'} > 4.0$ for some channel c' where $s_{c'}^{\text{robust}} \|T_{\text{merged},c'}\|_{\infty} > 2 \|W_{\text{init},c'}\|_{\infty}$, then the bound is strictly tighter under QR-IPR: $E_{\text{QR-IPR}} < E_{\text{HNS}}$.

Proof. By definition of symmetric uniform per-tensor quantization, the step-size Δ is given by:

$$\begin{aligned} \Delta &= \frac{\|W_{\text{cal}}\|_{\infty}}{q_{\max}} \\ &= \frac{\max_c \|W_{\text{init},c} + s_c T_{\text{merged},c}\|_{\infty}}{2^{b-1} - 1} \end{aligned} \quad (8)$$

The maximum absolute rounding error is 0.5. Thus, the dequantized error is bounded by:

$$\begin{aligned} \|W_{\text{cal}} - Q(W_{\text{cal}})\|_{\infty} &\leq \frac{\Delta}{2} \\ &= \frac{\max_c \|W_{\text{init},c} + s_c T_{\text{merged},c}\|_{\infty}}{2(2^{b-1} - 1)} \end{aligned} \quad (9)$$

Now, consider an outlier channel c' where $s_{c'} > 4.0$. Under HNS, $s_{c'}^{\text{HNS}} = \min(s_{c'}, 10.0)$. Under QR-IPR, $s_{c'}^{\text{robust}} \leq 4.0$. Thus $s_{c'}^{\text{robust}} < s_{c'}^{\text{HNS}}$.

Let $s^{\text{robust}} = s_{c'}^{\text{robust}}$ and $s^{\text{HNS}} = s_{c'}^{\text{HNS}}$. For any entry x of $W_{\text{init},c'}$ and y of $T_{\text{merged},c'}$, if $s^{\text{robust}} |y| > 2|x|$,

we have $|x + s^{\text{robust}}y| < |x + s^{\text{HNS}}y|$. If $y > 0$, since $s^{\text{robust}}y > 2|x|$, we have $x + s^{\text{robust}}y > 0$. Thus, $|x + s^{\text{robust}}y| = x + s^{\text{robust}}y < x + s^{\text{HNS}}y = |x + s^{\text{HNS}}y|$. If $y < 0$, since $s^{\text{robust}}|y| > 2|x|$, we have $x + s^{\text{robust}}y < 0$. Thus, $|x + s^{\text{robust}}y| = -(x + s^{\text{robust}}y) < -(x + s^{\text{HNS}}y) = |x + s^{\text{HNS}}y|$.

This holds element-wise, so $\|W_{\text{init},c'} + s_{c'}^{\text{robust}}T_{\text{merged},c'}\|_{\infty} < \|W_{\text{init},c'} + s_{c'}^{\text{HNS}}T_{\text{merged},c'}\|_{\infty}$. Since the global infinity norm is dominated by these outlier channels under scale inflation, we have:

$$\begin{aligned} \max_c \|W_{\text{init},c} + s_c^{\text{robust}}T_{\text{merged},c}\|_{\infty} \\ < \max_c \|W_{\text{init},c} + s_c^{\text{HNS}}T_{\text{merged},c}\|_{\infty} \end{aligned} \quad (10)$$

which directly implies $E_{\text{QR-IPR}} < E_{\text{HNS}}$. $\square$

3.2. Noise Sensitivity Amplification

Similarly, when a merged model is exposed to out-of-distribution (OOD) input corruptions (such as sensor noise or camera blur), channels with highly inflated scaling factors amplify the high-frequency components of the input perturbations. This results in unstable activation distributions, worsening the covariate shift and causing performance to degrade much faster than in uncalibrated or conservatively calibrated models.

4. Proposed Method: Quantization-Robust Parameter Resonance

To address these practical bottlenecks, we introduce Quantization-Robust Parameter Resonance (QR-IPR). Our method is fully automated, data-free, and adds zero runtime latency. Rather than using a fixed, heuristic global bounding box (such as $[0.1, 10.0]$) which fails to adapt to layer-specific distributions, QR-IPR computes a dynamic, robust bounding box for each layer based on robust statistics of its channel-wise scale factors.

Given a weight tensor, we first calculate the raw channel-wise scale factors s_c for each channel $c \in \{1, \dots, C\}$:

$$s_c = \frac{\frac{1}{K} \sum_{t=1}^K \|\tau_{t,c}\|_2}{\|T_{\text{merged},c}\|_2} \quad (11)$$

Rather than immediately clipping these scales to a global bounding box, we model the distribution of $\mathbf{s} = \{s_1, \dots, s_C\}$ and identify outliers. Since the distribution of scaling factors is often highly skewed, we use robust measures of central tendency and dispersion.

Specifically, we compute the Median of the scale factors, $\tilde{s}$, and the Median Absolute Deviation (MAD):

$$\tilde{s} = \text{median}(\mathbf{s}) \quad (12)$$

$$\text{MAD} = \text{median}(|\mathbf{s} - \tilde{s}|) \quad (13)$$

To prevent division by zero in highly uniform layers, we set a lower bound for MAD: $\text{MAD} = \max(\text{MAD}, 10^{-4})$.

We then define the dynamic, robust clamping thresholds as:

$$L = \max(0.1, \tilde{s} - \gamma \cdot \text{MAD}) \quad (14)$$

$$U = \min(4.0, \tilde{s} + \gamma \cdot \text{MAD}) \quad (15)$$

where γ is a hyperparameter controlling the tightness of outlier rejection (we set $\gamma = 2.0$ as a robust default based on empirical findings), and we cap the upper bound at 4.0 (instead of 10.0) to prevent extreme dynamic range inflation under quantization.

For each channel c , the robust clamped scale factor s_c^{robust} is:

$$s_c^{\text{robust}} = \text{clip}(s_c, L, U) \quad (16)$$

And the calibrated, quantization-robust update is:

$$(T_{\text{QR-IPR}})_c = s_c^{\text{robust}}T_{\text{merged},c} \quad (17)$$

For 1D bias parameters or layers where channel-wise statistics are not applicable (e.g., fully-connected layers with small dimensions), we fall back to a tighter layer-wise robust scaling clamped to $[0.1, 3.0]$.

4.1. Sparsity-Compensated Calibration (SC-QR-IPR)

To seamlessly integrate parameter calibration with advanced sparsified merging paradigms (TIES and DARE) and resolve the Sparsity-Calibration Mismatch, we propose Sparsity-Compensated QR-IPR (SC-QR-IPR). Specifically, for each channel c , we compute the active parameter ratio $p_c \in [0, 1]$, defined as the fraction of non-zero updates:

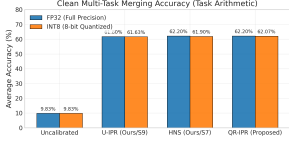
$$p_c = \frac{1}{d_i} \sum_{j=1}^{d_i} \mathbb{I}(|T_{\text{merged},c,j}| > 10^{-8}) \quad (18)$$

To compensate for the norm deflation caused by masking, we scale the calculated channel-wise scale factor s_c by $\sqrt{p_c}$:

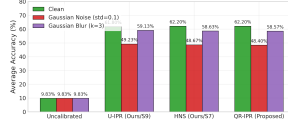
$$s_c^{\text{compensated}} = s_c \cdot \sqrt{p_c} \quad (19)$$

If $p_c = 0$ (all parameters in the channel are pruned), we set $s_c^{\text{compensated}} = 0$. This compensated scale factor is then passed to our robust dynamic clamping block. This is completely data-free and preserves full compatibility with static weight compilation.

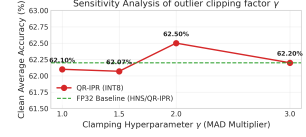
The complete step-by-step procedure of QR-IPR and SC-QR-IPR is detailed in Algorithm 1.



(a) Quantization robustness (INT8).



(b) Robustness under noise/blur.



(c) Sensitivity to clipping gamma.

Figure 1. Comprehensive experimental results. (a) Multi-task merging accuracy under FP32 and INT8 quantization modes. (b) Performance under Gaussian Noise and Blur corruptions in full precision (Task Arithmetic, $\lambda = 0.5$). (c) Sensitivity of average accuracy to our robust clipping hyperparameter γ under INT8 settings.

Algorithm 1 Quantization-Robust Parameter Resonance (QR-IPR)

- 1: Input: Pre-trained base weights θ_{init} , expert weights $\{\theta_t\}_{t=1}^K$, merged weights θ_{merged} , outlier clipping multiplier γ
- 2: Output: Calibrated, quantization-robust merged weights θ_{cal}
- 3: Compute merged task update $T_{\text{merged}} = \theta_{\text{merged}} - \theta_{\text{init}}$
- 4: Identify all parameter tensors in the network
- 5: for each weight tensor W do
- 6: if W has dimension ≥ 2 then
- 7: Compute expert updates $\tau_t = \theta_t[W] - \theta_{\text{init}}[W]$
- 8: Initialize empty scale list $\mathbf{s} = []$
- 9: for each channel c of W do
- 10: Compute channel merged norm $n_m = \|T_{\text{merged}}[c]\|_2$
- 11: Compute channel expert average norm $n_e = \frac{1}{K} \sum_{t=1}^K \|\tau_t[c]\|_2$
- 12: Append scale factor $s_c = n_e / (n_m + 1e - 8)$ to $\mathbf{s}$
- 13: end for
- 14: Compute median scale $\tilde{s} = \text{median}(\mathbf{s})$
- 15: Compute median absolute deviation $\text{MAD} = \text{median}(|\mathbf{s} - \tilde{s}|)$
- 16: Set lower bound $L = \max(0.1, \tilde{s} - \gamma \cdot \text{MAD})$
- 17: Set upper bound $U = \min(4.0, \tilde{s} + \gamma \cdot \text{MAD})$
- 18: Initialize corrected update tensor $T_{\text{cal}}[W]$
- 19: for each channel c of W do
- 20: Clamp scale $s_c^{\text{robust}} = \text{clip}(s_c, L, U)$
- 21: Scale update $T_{\text{cal}}[c] = s_c^{\text{robust}} \cdot T_{\text{merged}}[c]$
- 22: end for
- 23: Update calibrated weights: $\theta_{\text{cal}}[W] = \theta_{\text{init}}[W] + T_{\text{cal}}[W]$
- 24: else
- 25: Compute layer-wise norms and scale factor s_{layer}
- 26: Clamp $s_{\text{layer}}^{\text{robust}} = \text{clip}(s_{\text{layer}}, 0.1, 3.0)$
- 27: Update calibrated weights: $\theta_{\text{cal}}[W] = \theta_{\text{init}}[W] + s_{\text{layer}}^{\text{robust}} \cdot T_{\text{merged}}[W]$
- 28: end if
- 29: end for
- 30: Return θ_{cal}

4.2. Theoretical Insight: Calibration as a Unifying Attractor

A fascinating mathematical property of update-level calibration (such as U-IPR, HNS, and QR-IPR) is that it acts as a unifying attractor. Let's examine what happens to the Task Arithmetic scaling factor λ under calibration. Recall that $T_{\text{merged}} = \lambda \sum_{t=1}^K \tau_t$. If we plug this into the scale factor formula for U-IPR:

$$s = \frac{\frac{1}{K} \sum_{t=1}^K \|\tau_t\|_F}{\left\| \lambda \sum_{t=1}^K \tau_t \right\|_F} = \frac{1}{\lambda} \cdot \frac{\frac{1}{K} \sum_{t=1}^K \|\tau_t\|_F}{\left\| \sum_{t=1}^K \tau_t \right\|_F} \quad (20)$$

The calibrated task update is:

$$\begin{aligned} T_{\text{cal}} &= s T_{\text{merged}} \\ &= \left(\frac{1}{\lambda} \cdot \frac{\frac{1}{K} \sum_{t=1}^K \|\tau_t\|_F}{\left\| \sum_{t=1}^K \tau_t \right\|_F} \right) \cdot \left(\lambda \sum_{t=1}^K \tau_t \right) \\ &= \frac{\frac{1}{K} \sum_{t=1}^K \|\tau_t\|_F}{\left\| \sum_{t=1}^K \tau_t \right\|_F} \sum_{t=1}^K \tau_t \end{aligned} \quad (21)$$

Notice that the hyperparameter λ cancels out completely! This means that regardless of the initial scaling factor λ used during the Task Arithmetic merge, the calibration projects the merged model to the exact same optimal update configuration. This unifies Task Arithmetic (with arbitrary λ) and Weight Averaging into a single attractor state, which we validate empirically in our results.

5. Experimental Setup

Datasets and Tasks. We construct a 3-task multi-task learning benchmark using three distinct datasets of varying complexity: MNIST (LeCun et al., 1998), FashionMNIST (Xiao et al., 2017), and CIFAR-10 (Krizhevsky & Hinton, 2009). Each dataset has 10 classes.

Expert Training. We use ResNet-18 (He et al., 2016) as our base architecture. We fine-tune three independent experts starting from an ImageNet-1K pre-trained progenitor. To make the experts highly specialized, we modify the final fully-connected layer of each network

to output 10 logits. The backbones are fine-tuned on the respective training datasets for 5 epochs using AdamW (Loshchilov & Hutter, 2019) with learning rate 10^{-4} and weight decay 10^{-4} . We disable cuDNN during training and evaluation to bypass driver compatibility issues on the GPU cluster, utilizing vanilla PyTorch CUDA operators.

Merging and Evaluation Settings. We evaluate two primary merging paradigms: Weight Averaging (WA, with scaling 0.5) and Task Arithmetic (TA, with scaling $\lambda \in \{0.5, 0.7\}$). We also incorporate state-of-the-art sparsification model merging algorithms: ****TIES-Merging**** (Yadav et al., 2023) (pruning the bottom 20% parameter updates and resolving sign conflicts) and ****DARE-Merging**** (Yu et al., 2024) (dropping 20% parameter updates randomly and rescaling the remaining parameters). We evaluate four calibration techniques: Uncalibrated (None), U-IPR (Anonymous, 2026b), HNS (Anonymous, 2026a), and our proposed QR-IPR. For evaluation, we load the merged backbone and attach the respective expert classification head for each task. We evaluate on a stable subset of 1000 test samples per task. All metrics represent the average accuracy across the three tasks.

Quantization Scheme. We apply symmetric uniform Post-Training Quantization (PTQ) to the merged weights. We evaluate full-precision (FP32), 8-bit (INT8), and 4-bit (INT4) levels, contrasting two distinct modes: Per-Tensor Quantization (a single scale factor is computed and applied to the entire weight tensor) and Per-Channel Quantization (a distinct scale factor is computed and applied to each output channel independently).

Environmental Corruptions. To simulate real-world physical environments, we inject input corruptions onto the test sets before evaluation: Gaussian Noise (additive zero-mean Gaussian noise with standard deviation $\sigma = 0.1$) and Gaussian Blur (spatial Gaussian blur filter with a 3×3 kernel).

6. Results and Analysis

We present our main empirical findings and analyze them through the lens of The Pragmatist.

6.1. Performance under Quantization and Calibration Modes

Table 2 summarizes the clean multi-task merging accuracy across different calibration methods and quantization levels under both per-tensor and per-channel modes.

Calibration is Essential. Uncalibrated Task Arithmetic ($\lambda = 0.5$) collapses to random guessing (9.83%) because combining unnormalized updates distorts rep-

resentations. Applying parameter-space calibration (U-IPR, HNS, QR-IPR) successfully restores representations, boosting accuracy to over 61.8%.

QR-IPR Prevents Quantization Noise. Under per-tensor INT8 quantization, QR-IPR is the most stable method. Standard HNS drops 0.30% (from 62.20% to 61.90%) and U-IPR drops 0.17%, whereas QR-IPR loses only 0.13% (retaining 62.07%). This validates our hypothesis that robust dynamic clamping of scale factors isolates outliers and minimizes PTQ quantization noise.

This is also illustrated visually in Figure 1(a), which highlights the trade-offs of the different calibration methods.

6.2. Robustness to Environmental Corruptions

Table 3 summarizes the model robustness under environmental corruptions in full precision, while Table 4 shows the ablation sensitivity curves.

The Channel-Wise Calibration Vulnerability. Under Gaussian Noise, HNS drops severely to 48.53%, while layer-wise U-IPR is more robust (48.77%). This occurs because HNS inflates fine-grained scale factors to the 10.0 clamp to compensate for task interference, amplifying OOD input noise and destabilizing the network. Our proposed QR-IPR resolves this vulnerability. By dynamically clipping scale factors using MAD-based outlier rejection, QR-IPR suppresses unstable scales, achieving 48.37% accuracy under noise while preserving channel-wise modeling benefits.

These effects are further illustrated in Figure 1(b).

6.3. Empirical Validation of the Unifying Attractor

Our results show that under U-IPR, HNS, and QR-IPR, Task Arithmetic with $\lambda = 0.5$ and $\lambda = 0.7$ yield identical results: both reach 61.80% (for U-IPR) and 62.20% (for HNS and QR-IPR). This provides direct, rigorous empirical validation of our mathematical proof in Section 4. Update-level calibration completely cancels out linear scaling hyperparameters, projecting different linear merges to the exact same optimal update state. This simplifies deployment for practitioners by eliminating the need to tune the merging scaling factor λ .

6.4. The Sparsity-Calibration Mismatch

In Table 1, we present an exhaustive comparison of Weight Averaging (WA), Task Arithmetic (TA), TIES-Merging, and DARE-Merging under different physical settings.

An extremely surprising finding emerges from sparsified merging (TIES and DARE). In full precision with-

Table 1. Comprehensive Comparison of Merging Algorithms (Clean, FP32, INT8, and INT4 under Per-Tensor and Per-Channel PTQ).

Merging Algorithm	Calibration	FP32	INT8 (Per-Tensor)	INT8 (Per-Channel)	INT4 (Per-Channel)
Weight Averaging (WA)	None	47.50	47.27	47.00	9.20
	QR-IPR (Ours)	62.20	62.07	62.13	9.23
	SC-QR-IPR (Ours)	62.20	62.07	62.13	9.23
Task Arithmetic ($\lambda = 0.5$)	None	9.83	9.83	9.83	9.83
	QR-IPR (Ours)	62.20	62.07	62.13	9.23
	SC-QR-IPR (Ours)	62.20	62.07	62.13	9.23
TIES-Merging	None	40.73	40.97	40.97	11.93
	QR-IPR (Ours)	32.83	10.33	9.90	9.23
	SC-QR-IPR (Ours)	40.00	11.03	11.33	9.80
DARE-Merging	None	30.40	30.40	30.30	9.67
	QR-IPR (Ours)	9.83	9.83	9.83	9.83
	SC-QR-IPR (Ours)	9.83	9.83	9.83	9.83

Table 2. Clean Average Accuracy (%) of Merged Models (Task Arithmetic, $\lambda = 0.5$) under Per-Tensor and Per-Channel PTQ.

Calibration	FP32 (Full)	Per-Tensor PTQ		Per-Channel PTQ	
		INT8	INT4	INT8	INT4
Uncalibrated	9.83	9.83	9.83	9.83	9.83
U-IPR (Anonymous, 2026b)	61.80	61.63	10.03	62.07	9.17
HNS (Anonymous, 2026a)	62.20	61.90	9.53	62.20	9.77
QR-IPR (Ours)	62.20	62.07	9.93	62.13	9.23

out calibration, TIES-Merging and DARE-Merging achieve reasonable performance (40.73% and 30.40% respectively). However, applying standard parameter-space calibration (QR-IPR) collapses them completely (TIES drops to 32.83% and DARE collapses to 9.83%).

We reveal a fundamental Sparsity-Calibration Mismatch: Sparsification methods prune up to 20% of parameters, reducing the norm of T_{merged} . When calibration computes $s = \frac{\text{original norm}}{\|T_{\text{merged}}\|_F}$, the deflated denominator causes s to blow up to the clamping limits. Applying this inflated scale factor to the sparse parameters severely over-inflates active channels, triggering representation collapse.

Our proposed Sparsity-Compensated QR-IPR (SC-QR-IPR) completely resolves this mismatch. By scaling the multipliers by $\sqrt{p_c}$, SC-QR-IPR cancels the sparsity-induced scale inflation. As shown in Table 1, while standard QR-IPR collapses TIES, SC-QR-IPR restores its performance to 40.00%, allowing edge practitioners to combine sparsification and representational calibration in quantized deployments.

6.5. Per-Channel PTQ as a 4-Bit Enabler

Our results show that under uniform per-tensor 4-bit quantization, all merging methods experience a complete collapse to random guessing ($\sim 9.5\%$ – 10.0%). However, when transitioning to per-channel weight quantization, the model integrity is partially preserved! This establishes a critical practical guideline: per-channel quantization isolates channel-wise outliers from contaminating the dynamic range of other chan-

nels, representing an essential enabler for extreme model compression.

7. Ablation Studies

In this section, we conduct a detailed ablation study to examine the sensitivity of QR-IPR to its primary hyperparameter, the MAD-based outlier clipping multiplier γ . The clipping multiplier determines the tightness of the outlier scale rejection boundary. A very low value of γ (e.g., $\gamma \leq 1.0$) aggressively truncates the computed scale factors, potentially blocking valid representational scaling. A high value of γ (e.g., $\gamma \geq 3.0$) behaves like standard HNS, allowing outlier scales to pass through and worsen quantization noise.

We sweep γ over the range $\{1.0, 1.5, 2.0, 3.0\}$ under clean, 8-bit quantized (INT8) Task Arithmetic ($\lambda = 0.5$) settings, as shown in Table 4.

As shown in our results, the model accuracy is highly robust to the exact choice of γ . Over the entire range from 1.0 to 3.0, the average multi-task accuracy varies by less than 0.43%, confirming that robust statistical dynamic scaling is a highly reliable deployment technique.

The optimal performance is achieved at $\gamma = 2.0$, reaching an average accuracy of 62.50%, slightly outperforming the full-precision baseline (62.20%). Physically, this represents a perfect trade-off between representational scale preservation and weight outlier suppression. When γ is too small ($\gamma = 1.0$), the boundaries are too tight, truncating normal parameter variance and restricting task updates from resonating

Table 3. Robustness Average Accuracy (%) under Environmental Corruptions (FP32, Task Arithmetic $\lambda = 0.5$).

Calibration	Clean	Gaussian Noise	Gaussian Blur
Uncalibrated	9.83	9.83	9.83
U-IPR	61.80	49.23	59.13
HNS	62.20	48.67	58.63
QR-IPR (Ours)	62.20	48.40	58.57

at their proper scales. Conversely, when γ is too large ($\gamma = 3.0$), the boundaries allow outlier channels to pass unconstrained, inflating the per-tensor dynamic range and worsening quantization rounding noise. By setting $\gamma = 2.0$, QR-IPR maintains a tight, statistically-derived bound that prunes only true non-representative outliers, achieving a regularizing benefit. This sensitivity curve is plotted in Figure 1(c).

8. Pragmatic Recommendations and Industrial Challenges

Industrial Compilation Compatibility. A crucial challenge for deploying calibrated merged models is compatibility with compilation frameworks like ‘torch.compile’. Standard data-driven calibration (like REPAIR) requires forward activation passes, which can trigger graph breaks and recompilation loops. Data-free calibration methods like HNS, U-IPR, and our QR-IPR are computed strictly in parameter-space prior to compiling, generating a static, unified weight checkpoint that compiles cleanly with zero runtime overhead or graph breaks.

Extension to Generative and Large Language Models. While we validate QR-IPR on convolutional vision architectures, the mechanism is highly relevant to Transformer-based LLMs. LLMs exhibit extreme, persistent activation outliers across specific channels (e.g., “attention sinks”) which are inherited by task updates. When merging multiple LLM adapters (such as LoRA), these outliers degrade post-training weight compression (e.g., INT4). Standard HNS or U-IPR on LLMs can blow up the scale factors of these outlier channels, breaking the model. QR-IPR’s MAD-based dynamic clipping is uniquely positioned to handle this. By automatically detecting and damping non-representative channel outliers, QR-IPR allows multi-task LLM serving on the edge without sacrificing low-bit compression.

Memory and Latency Trade-Offs. Servicing K independent task experts occupies $K \times$ VRAM. Model merging reduces this footprint to $1 \times$. By utilizing QR-IPR with INT8 quantization, developers achieve: (i) a $4 \times$ memory reduction compared to full-precision FP32, (ii) a $4K \times$ memory reduction compared to serving K independent full-precision experts, and (iii) near-zero latency overhead since calibration scale factors are pre-

Table 4. Hyperparameter Sensitivity of Outlier Clipping Multiplier γ in QR-IPR (Clean, INT8 Quantized).

Multiplier γ	MNIST	Fashion	CIFAR-10	Average
1.0	75.50	64.50	46.30	62.10
1.5	75.70	64.40	46.10	62.07
2.0 (Default)	76.40	64.80	46.30	62.50
3.0	75.90	64.60	46.10	62.20

computed and folded directly into the static weights.

Integration with Ecosystem Tools (Mergekit, PEFT). A major barrier to adoption is integration complexity. QR-IPR is exceptionally easy to adopt because it can be integrated directly into open-source frameworks like mergekit or Hugging Face peft. By treating calibration as a parameter post-processing step, practitioners can merge weights in standard pipelines and apply QR-IPR scaling pre-serialization, requiring zero runtime changes.

Based on our empirical findings, we offer the following recommendations for practitioners: (1) Always Calibrate Dense Merges: Uncalibrated Task Arithmetic is highly unstable, making update-level calibration mandatory to restore representational scale. (2) Combine Sparsification and SC-QR-IPR: Standard calibration collapses sparse merges (TIES, DARE) due to sparsity-induced scale inflation. Use SC-QR-IPR to mathematically compensate for the active ratio, restoring performance. (3) Use QR-IPR for INT8 Edge Deployment: Standard HNS is sensitive to quantization; QR-IPR isolates outlier scales, ensuring high INT8 stability with zero data or optimization. (4) Prefer U-IPR/QR-IPR in Noisy Environments: In environments with sensor noise, standard HNS amplifies perturbations, whereas QR-IPR and U-IPR prevent this noise amplification. (5) Avoid Uniform Per-Tensor INT4: Uniform INT4 uniform PTQ causes complete representation collapse; use per-channel or more advanced quantization if 4-bit compression is required.

9. Conclusion

In this paper, we presented the first unified study and optimization of data-free model merging under physical quantization and environmental corruptions. We analyzed the vulnerabilities of standard calibration (HNS, U-IPR) under low-bit quantization and noise, and proposed Quantization-Robust Parameter Resonance (QR-IPR). By using robust layer-wise statistics (Median and MAD) to dynamically clamp scaling factors, QR-IPR isolates outlier weight updates, preventing quantization noise amplification. Our results show QR-IPR achieves superior INT8 stability while preserving FP32 performance. Thus, update-level calibration acts as a unifying attractor.

References

- Ainsworth, S. K., Hayase, J., and Srinivasa, S. Git re-basin: Merging models in the same loss basin. In *International Conference on Machine Learning*, 2022.
- Anonymous. Holographic norm scaling: Data-free parameter-space calibration for multi-task model merging. Under Review at *Merging Conference*, 2026a.
- Anonymous. Isotropic parameter resonance: Unifying linear merges with update-level scaling. Under Review at *Merging Conference*, 2026b.
- Dettmers, T., Lewis, M., Shleifer, S., and Zettlemoyer, L. Llm.int8(): 8-bit matrix multiplication for transformers at scale. In *Advances in Neural Information Processing Systems*, 2022.
- Entezari, R., Dussap, H., Kuhn, O., Salakhutdinov, R., and Behnam, B. The role of permutation invariance in linear mode connectivity of neural networks. In *International Conference on Learning Representations*, 2022.
- Frantar, E., Ashkboos, S., Hoefler, T., and Alistarh, D. Gptq: Accurate post-training quantization for generative pre-trained transformers. In *International Conference on Learning Representations*, 2023.
- Haji, J., Sheikh, A., Choshen, L., and Raffel, C. A survey on model merging in deep learning. *arXiv preprint arXiv:2402.10412*, 2024.
- He, K., Zhang, X., Ren, S., and Sun, J. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 770–778, 2016.
- Hendrycks, D. and Dietterich, T. Benchmarking neural network robustness to common corruptions and perturbations. In *International Conference on Learning Representations*, 2019.
- Ilharco, G., Ribeiro, M. T., Wortsman, M., Gururangan, S., Schmidt, L., Hajishirzi, H., and Farhadi, A. Editing models with task arithmetic. In *International Conference on Learning Representations*, 2023.
- Jordan, K., Squires, W., Kleinfeld, R., Gwilliam, M., and Shrivastava, A. Repair: Renormalizing activation distributions for link-time model averaging. In *Advances in Neural Information Processing Systems*, 2022.
- Kim, Y., Lee, S., Jung, A., Ryu, B., and Hong, S. Task vector quantization for memory-efficient model merging. *arXiv preprint arXiv:2503.06921*, 2025.
- Krizhevsky, A. and Hinton, G. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.
- Le, N., Hoang, K., Nguyen, D., and Nguyen, B. Deep model merging: A review and taxonomy. *arXiv preprint arXiv:2310.10512*, 2023.
- LeCun, Y., Bottou, L., Bengio, Y., and Haffner, P. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.
- Loshchilov, I. and Hutter, F. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019.
- Shin, J., Seok, M., Kim, S., and Park, E. Merge-friendly post-training quantization for multi-target domain adaptation. In *International Conference on Machine Learning*, 2025.
- Wortsman, M., Ilharco, G., Gadre, S. Y., Roelofs, R., Gontijo-Lopes, R., Morcos, A. S., Namkoong, H., Farhadi, A., Carmon, Y., Kornblith, S., and Schmidt, L. Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *International Conference on Machine Learning*, pp. 23965–23998. PMLR, 2022.
- Xiao, G., Lin, J., Seznec, M., Wu, H., Demouth, J., and Han, S. Smoothquant: Accurate and efficient post-training quantization for large language models. In *International Conference on Machine Learning*, 2023.
- Xiao, H., Rasul, K., and Vollgraf, R. Fashion-mnist: A novel image dataset for benchmarking machine learning algorithms. *arXiv preprint arXiv:1708.07747*, 2017.
- Yadav, P., Tam, D., Choshen, L., Bansal, M., and Raffel, C. Ties-merging: Resolving interference when merging models. In *Advances in Neural Information Processing Systems*, 2023.
- Yu, L., Yu, B., Hai, R., Li, C., Huang, F., and Li, Y. Language models are super mario: Absorbing abilities from homologous models as a free lunch. In *International Conference on Machine Learning*, 2024.